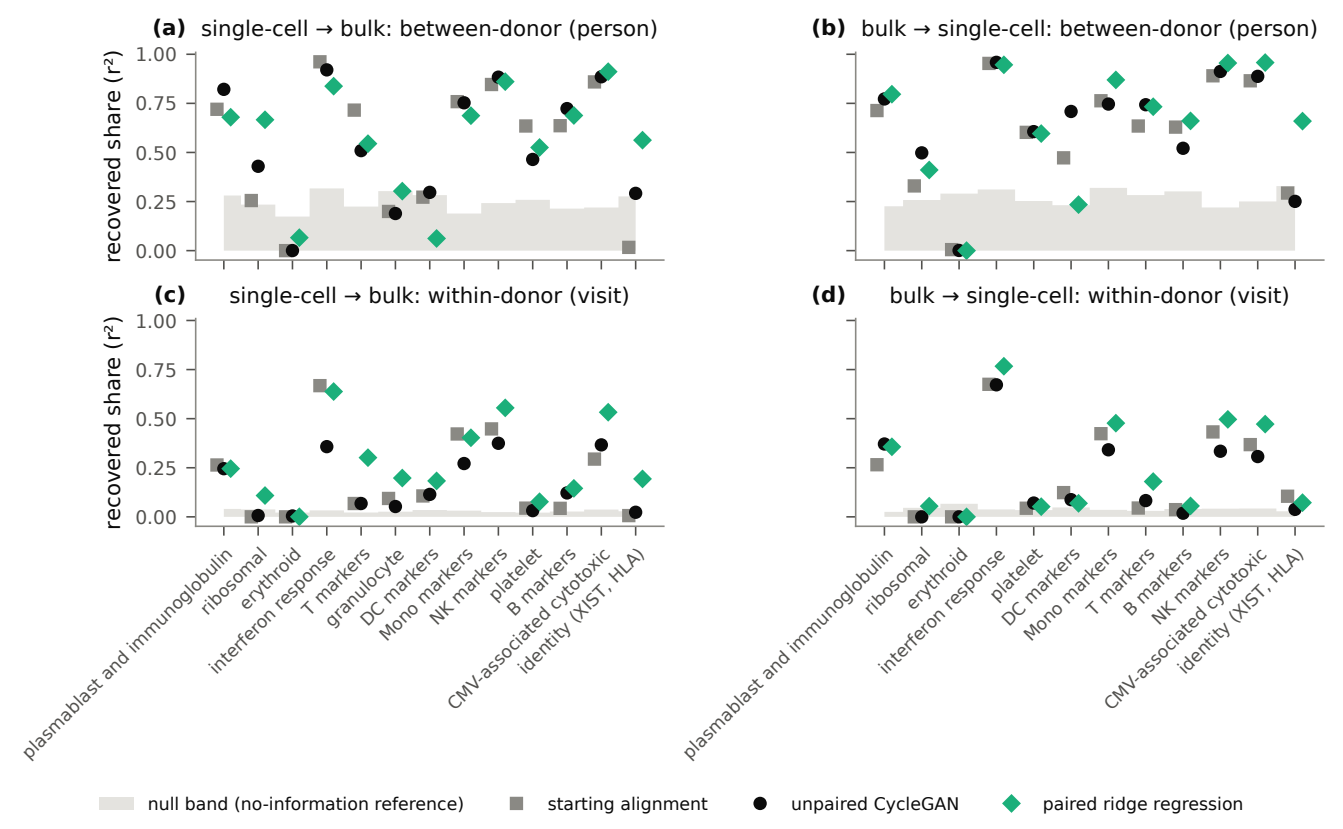
(a) single-cell → bulk: between-donor (person)
(b) bulk → single-cell: between-donor (person)
(c) single-cell → bulk: within-donor (visit)
(d) bulk → single-cell: within-donor (visit)
recovered share (r²)
1.00
0.75
0.50
0.25
0.00
plasmablast and immunoglobulin
ribosomal
erythroid
interferon response
T markers
granulocyte
DC markers
Mono markers
NK markers
platelet
B markers
CMV-associated cytotoxic
identity (XIST, HLA)
null band (no-information reference)
starting alignment
unpaired CycleGAN
paired ridge regression